Short Communication

Selective Decision Intelligence Under Predictive Uncertainty: Calibration, Human Deferral, and Error Concentration in AI Automation

Srihitha Reddy Betelly[a,*]

[a]*Midwest University, Saint Louis, MO, USA*

*Corresponding author. E-mail: srihithareddy.bethelly@gmail.com

Abstract

Data-driven artificial intelligence increasingly supports operational and managerial decisions, yet predictive confidence is often treated as if it were a reliable estimate of correctness. This short communication proposes and evaluates an Uncertainty-Calibrated Decision Intelligence (UCDI) framework that combines predictive modelling, post-hoc probability calibration, uncertainty-ranked selective prediction and an explicit human-deferral rule. Experiments use a public credit-default benchmark of 30,000 clients and 23 predictors with non-overlapping training (18,000), calibration (6,000) and test (6,000) partitions of identical default prevalence (22.12%). Logistic regression, random forest and gradient boosting were compared in raw, sigmoid and isotonic variants. Gradient boosting gave the best discrimination (ROC-AUC 0.7861), whereas calibration effects were model-dependent: isotonic regression reduced the expected calibration error of logistic regression from 0.0553 to 0.0161 but lowered its error capture under deferral. With the decision model selected on the calibration split, automating the 80% most confident test cases raised accuracy from 0.8202 to 0.8717 (a 28.6% relative risk reduction), while the 20% deferred to human review contained 42.91% of all model errors, a 2.15× lift over random deferral (permutation $p < 0.001$). Across ten independent splits, selective accuracy was 0.8692 ± 0.0031 and lift 2.08 ± 0.04. The results support uncertainty-guided deferral as a reproducible decision-control mechanism and show that calibration should be validated per model, and for the decision it serves, rather than assumed to improve reliability.

Keywords: Decision intelligence; Selective prediction; Probability calibration; Human–AI collaboration; Knowledge-based decision support; Digital transformation

1. Introduction

Organisations pursuing digital transformation are moving AI from analytical dashboards into the decisions themselves: approving credit, routing claims and flagging transactions [1–3]. Firm-level studies link this shift to performance only when AI capability is matched by governance and complementary human judgement [4–6]. In practice, the key managerial question is often not *whether* to use a model but *which* decisions it should be trusted to make alone, and which should be escalated to a person [7, 8].

Answering that question requires predicted probabilities that mean what they say. Uncertainty quantification and explainability surveys repeatedly warn that raw classifier scores are not reliable estimates of correctness [9, 10]. Credit scoring illustrates the stakes: benchmark studies compare dozens of classifiers on discrimination [11–13]; recent *Knowledge-Based Systems* contributions improve ensembles and deep architectures for credit risk [14–16]; and explainable credit-evaluation frameworks have been proposed [17]. These works mostly optimise accuracy over *all* applicants. Classical reject-option theory [18] and its modern learning-to-reject extensions [19, 20] instead let a model abstain, and profit-oriented uncertainty estimation has recently been proposed for credit scoring [21]. Khalili et al. embed calibration, cross-conformal uncertainty and governance in one credit-scoring architecture [22],

and a 36-dataset tabular benchmark finds that the best calibrator depends on the model family [23]. What remains under-examined is how the *choice of calibration method* interacts with the *quality of deferral*, that is, the share of errors routed to human review, in a business decision pipeline.

This short communication addresses that gap with three contributions. (i) It formalises UCDI, a lightweight knowledge-based decision layer that turns calibrated probabilities into an automate-or-defer rule with an explicit coverage target. (ii) It reports a leakage-free experiment with nine model–calibration variants, a 1,000-trial random-deferral baseline and ten-split robustness analysis. (iii) It shows empirically that better calibration error does not imply better deferral, which matters for managers who adopt calibration as a default governance step.

2. The UCDI framework

Let f be a trained classifier and g a calibration map fitted on a held-out calibration set, giving calibrated default probabilities $p(x) = g(f(x))$. The automated decision is $\hat{y} = 1[p(x) \geq 0.5]$. UCDI scores each case by its decision uncertainty

$$u(x) = 1 - |2p(x) - 1|, \tag{1}$$

which equals 0 for a fully confident prediction and 1 at the decision boundary. For a target automation coverage $c \in (0, 1]$, the automated set A_c contains the $\lceil cn \rceil$ test cases with the lowest $u(x)$; all others are deferred to a human reviewer. Writing $e_i = 1[\hat{y}_i \neq y_i]$, the selective risk, error capture and lift are

$$R(c) = (1/|A_c|)\, \Sigma_{i \in A_c}\, e_i, \quad E(c) = \Sigma_{i \notin A_c}\, e_i \,/\, \Sigma_i\, e_i, \quad L(c) = E(c) \,/\, (1 - c). \tag{2}$$

$R(c)$ is the error rate the organisation accepts on automated decisions, $E(c)$ is the share of all model errors routed to human review, and $L(c)$ compares that share with a random reviewer allocation of the same size, for which $E(c) \approx 1 - c$ and $L \approx 1$. Calibration quality is measured by the expected calibration error over $B = 15$ equal-width bins, $\text{ECE} = \Sigma_b\, (n_b/n)|\bar{y}_b - \bar{p}_b|$, together with the Brier score and log-loss. Because a monotone calibration map preserves the ranking of raw scores, its effect on deferral comes from relocating the 0.5 decision point and, for isotonic regression, from the ties created by its step function. Calibration can therefore change $E(c)$ in either direction, which motivates testing it rather than assuming it helps.

3. Experimental design

The Taiwanese credit-card default data set [24] contains 30,000 clients, 23 predictors (credit limit, demographics, six months of repayment status, bill and payment amounts) and a binary next-month default label. It was split once, stratified, into training (60%), calibration (20%) and test (20%) partitions with prevalences of 22.12%, 22.12% and 22.12%. Three widely benchmarked families [11, 25] were trained: standardised logistic regression, a random forest (400 trees, minimum leaf size 5) and histogram gradient boosting (learning rate 0.05, 400 iterations, 31 leaves, L2 = 1.0). Each base model was used raw and with sigmoid (Platt) and isotonic calibration fitted only on the calibration partition.

To avoid test leakage, the decision model was chosen on the calibration partition as the raw model with the lowest Brier score (random forest), and isotonic calibration was applied as the UCDI default. The main operating point is $c = 0.80$. Significance of error capture was assessed by drawing 1,000 random deferral sets of the same size and computing the empirical p-value $(1 + \#\{\text{random} \geq \text{observed}\})/(1 + 1{,}000)$. Robustness was assessed by repeating the full pipeline on ten further stratified splits (seeds 100–109). All code (Python 3.13, scikit-learn 1.9) and outputs are in a public repository (see Data availability).

4. Results

Table 1 reports test-set performance for the nine variants. Gradient boosting gave the highest ROC-AUC (0.7861) and PR-AUC (0.5667), with the random forest close behind (0.7807), and both clearly ahead of logistic regression (0.7262). Calibration effects were model-dependent (Fig. 1), consistent with benchmark evidence of a model–calibrator interaction [23]. Isotonic regression cut the ECE of logistic regression from 0.0553 to 0.0161 and improved its Brier score, but it *raised* the ECE of gradient boosting from 0.0152 to 0.0252, and sigmoid scaling did not reduce ECE for any model. The tree ensembles were already well calibrated in raw form (random forest ECE 0.0123). Isotonic calibration also lowered PR-AUC for every model because its piecewise-constant output creates ties in the ranking.

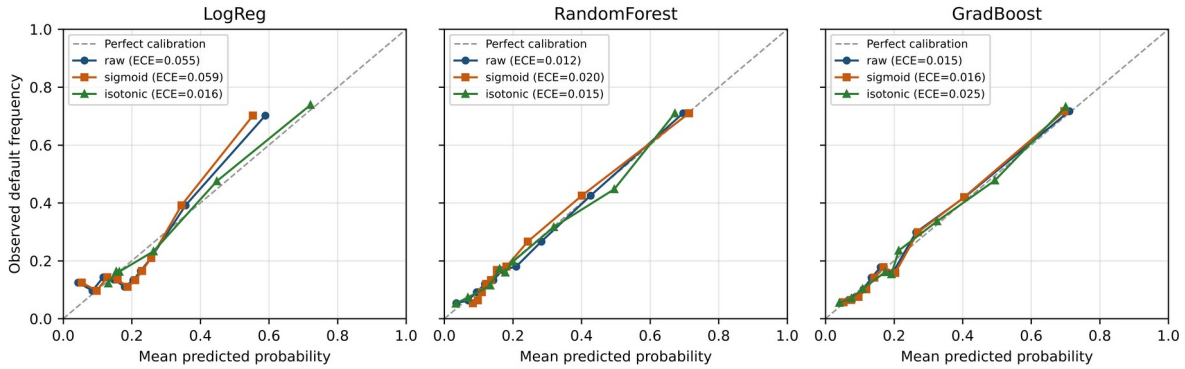


Fig. 1. Reliability diagrams on the test partition (n = 6,000, ten quantile bins) for raw, sigmoid and isotonic variants of each model. Legends give the expected calibration error (15 equal-width bins).

Table 1. Test-set discrimination, calibration and deferral quality of the nine model–calibration variants. $E(0.8)$ is the share of all errors captured by deferring the 20% most uncertain cases; Acc@0.8 is the accuracy on the 80% automated cases. Bold: UCDI decision model.

Model	Calibration	Accuracy	ROC-AUC	PR-AUC	Brier	ECE	E(0.8)	Acc@0.8
Logistic regression	Raw	0.8125	0.7262	0.5153	0.1436	0.0553	40.62%	0.8608
Logistic regression	Sigmoid	0.8087	0.7262	0.5153	0.1447	0.0591	41.46%	0.8600
Logistic regression	Isotonic	0.8203	0.7239	0.4935	0.1398	0.0161	37.11%	0.8588
Random forest	Raw	0.8197	0.7807	0.5647	0.1338	0.0123	42.88%	0.8712
Random forest	Sigmoid	0.8203	0.7807	0.5647	0.1341	0.0203	43.04%	0.8721
Random forest	**Isotonic**	**0.8202**	**0.7794**	**0.5417**	**0.1341**	**0.0150**	**42.91%**	**0.8717**
Gradient boosting	Raw	0.8227	0.7861	0.5667	0.1331	0.0152	41.35%	0.8700
Gradient boosting	Sigmoid	0.8225	0.7861	0.5667	0.1331	0.0165	41.60%	0.8704
Gradient boosting	Isotonic	0.8220	0.7835	0.5454	0.1340	0.0252	40.73%	0.8681

The most consequential pattern for decision design is that calibration error and deferral quality diverged. Isotonic logistic regression had one of the lowest ECEs yet the weakest error capture ($E(0.8)$ = 37.11% vs 40.62% raw), and gradient boosting, the best discriminator, captured fewer errors (41.35%) than the random forest (42.88%). Ranking models by accuracy or ECE alone would therefore have selected a weaker deferral policy.

Table 2 and Fig. 2 describe the UCDI operating curve for the selected model. At full automation the error rate is 0.1798. Automating 80% of cases reduces risk on the automated decisions to 0.1283 (accuracy 0.8717), a 28.6% relative reduction, while the 20% of deferred cases contain 42.91% of all errors against 20.02% for random deferral. None of the 1,000 random allocations matched this capture at any coverage level ($p < 0.001$). Lift is highest when reviewer capacity is scarce (2.52× at 5% deferral) and declines towards 1 as more cases are deferred, giving managers an explicit capacity–risk trade-off.

Table 2. UCDI risk–coverage profile for the decision model (random forest, isotonic) on the test partition. Random capture is the mean over 1,000 random deferral sets of equal size; p is the empirical permutation p-value. Bold: main operating point.

Coverage c	Deferred	Sel. accuracy	Sel. risk R(c)	Error capture E(c)	Random capture	Lift L(c)	p
1.000	0.000	0.8202	0.1798	—	—	—	—
0.950	0.050	0.8346	0.1654	12.60%	5.00%	2.52×	< 0.001
0.900	0.100	0.8491	0.1509	24.47%	10.00%	2.45×	< 0.001
0.850	0.150	0.8594	0.1406	33.55%	15.05%	2.24×	< 0.001
0.800	**0.200**	**0.8717**	**0.1283**	**42.91%**	**20.02%**	**2.15×**	**< 0.001**
0.700	0.300	0.8850	0.1150	55.24%	29.94%	1.84×	< 0.001
0.600	0.400	0.8972	0.1028	65.71%	40.07%	1.64×	< 0.001
0.500	0.500	0.9073	0.0927	74.24%	50.07%	1.48×	< 0.001

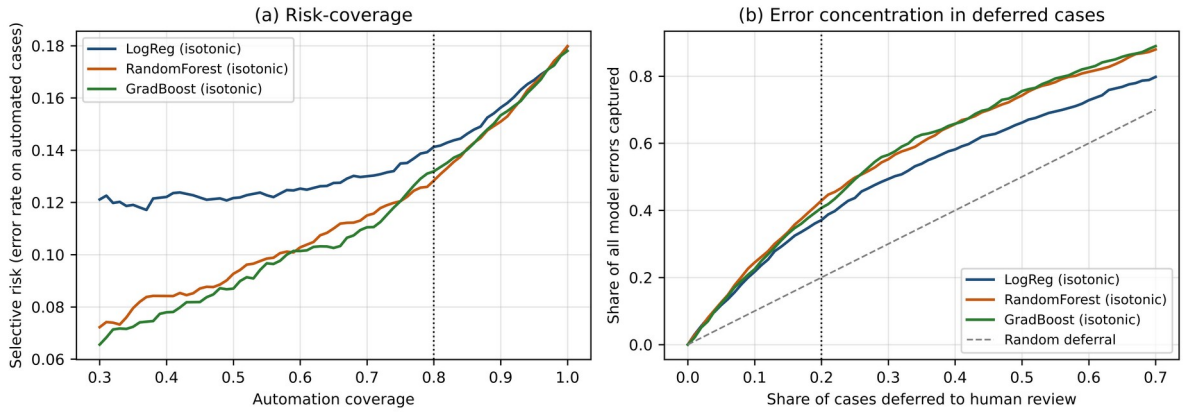


Fig. 2. (a) Risk–coverage curves of the isotonic-calibrated models; the dotted line marks $c = 0.80$. (b) Share of all model errors captured by deferring the most uncertain cases, compared with random deferral (dashed diagonal).

Robustness was high. Over ten independent splits, the selected model achieved selective accuracy 0.8692 ± 0.0031 at $c = 0.80$, error capture 41.62% and lift 2.08 ± 0.04, with every split above a 2.0× lift.

5. Discussion

For decision intelligence, the findings turn calibration and deferral from technical options into governance parameters. The coverage c is set by reviewer capacity and risk appetite, and UCDI makes the consequence of that choice measurable before deployment: by reviewing one case in five, the organisation can expect to intercept about four in ten model errors. This supports the human–AI symbiosis view of organisational decision making [7, 8] with a concrete, auditable allocation rule, and complements explainability-centred approaches [10, 17] by stating *when* explanation and review effort should be spent. The divergence between ECE and error capture also warns against treating calibration as a box-ticking step in AI governance: it should be validated per model, and against the decision metric it is meant to serve [9, 21, 23].

The study has limitations. It uses a single public data set from one market and period; the 0.5 threshold ignores the asymmetric costs of false approvals and false rejections; human reviewers are assumed to resolve deferred cases correctly, so the figures are upper bounds on system accuracy; and fairness of deferral across demographic groups was not assessed. Future work should combine cost-sensitive thresholds with profit-based uncertainty [21], model imperfect reviewers, and test the framework on further enterprise data sets.

6. Conclusion

UCDI shows that a simple, calibrated uncertainty gate can concentrate more than twice the random share of model errors into a small human-review queue while raising the accuracy of automated credit decisions by about five percentage points. Calibration helped some models and hurt others, so its benefit for deferral must be verified rather than assumed. The framework, code and data are released openly to support reproducible, knowledge-based decision support in digitally transforming organisations.

CRediT authorship contribution statement

Srihitha Reddy Betelly: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The data set is publicly available from the UCI Machine Learning Repository (https://doi.org/10.24432/C55S3H, CC BY 4.0). Code, the converted data file, all result tables and figure-generation scripts are available at https://github.com/Srihh21/ucdi-selective-decision-intelligence.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist with code development, drafting and language editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. All figures were generated programmatically from the experimental code; no generative AI image tools were used.

References

[1] Y. Duan, J.S. Edwards, Y.K. Dwivedi, Artificial intelligence for decision making in the era of Big Data – evolution, challenges and research agenda, International Journal of Information Management 48 (2019) 63–71. https://doi.org/10.1016/j.ijinfomgt.2019.01.021

[2] Y.K. Dwivedi, L. Hughes, E. Ismagilova, G. Aarts, C. Coombs, T. Crick, et al., Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy, International Journal of Information Management 57 (2021) 101994. https://doi.org/10.1016/j.ijinfomgt.2019.08.002

[3] G. Vial, Understanding digital transformation: A review and a research agenda, The Journal of Strategic Information Systems 28 (2) (2019) 118–144. https://doi.org/10.1016/j.jsis.2019.01.003

[4] P. Mikalef, M. Gupta, Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance, Information & Management 58 (3) (2021) 103434. https://doi.org/10.1016/j.im.2021.103434

[5] D. Wu, X. Lin, S. Gupta, A.K. Kar, Big Data Analytics Capability, Dynamic Capability, and Firm Performance: The Moderating Effect of IT–Business Strategic Alignment, IEEE Transactions on Engineering Management 71 (2024) 11638–11651. https://doi.org/10.1109/TEM.2024.3429648

[6] M. Kraus, S. Feuerriegel, A. Oztekin, Deep learning in business analytics and operations research: Models, applications and managerial implications, European Journal of Operational Research 281 (3) (2020) 628–641. https://doi.org/10.1016/j.ejor.2019.09.018

[7] M.H. Jarrahi, Artificial intelligence and the future of work: Human-AI symbiosis in organizational decision making, Business Horizons 61 (4) (2018) 577–586. https://doi.org/10.1016/j.bushor.2018.03.007

[8] X. Wu, L. Xiao, Y. Sun, J. Zhang, T. Ma, L. He, A survey of human-in-the-loop for machine learning, Future Generation Computer Systems 135 (2022) 364–381. https://doi.org/10.1016/j.future.2022.05.014

[9] M. Abdar, F. Pourpanah, S. Hussain, D. Rezazadegan, L. Liu, M. Ghavamzadeh, et al., A review of uncertainty quantification in deep learning: Techniques, applications and challenges, Information Fusion 76 (2021) 243–297. https://doi.org/10.1016/j.inffus.2021.05.008

[10] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, et al., Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI, Information Fusion 58 (2020) 82–115. https://doi.org/10.1016/j.inffus.2019.12.012

[11] S. Lessmann, B. Baesens, H.-V. Seow, L.C. Thomas, Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research, European Journal of Operational Research 247 (1) (2015) 124–136. https://doi.org/10.1016/j.ejor.2015.05.030

[12] B.R. Gunnarsson, S. vanden Broucke, B. Baesens, M. Óskarsdóttir, W. Lemahieu, Deep learning for credit scoring: Do or don't?, European Journal of Operational Research 295 (1) (2021) 292–305. https://doi.org/10.1016/j.ejor.2021.03.006

[13] E. Dumitrescu, S. Hué, C. Hurlin, S. Tokpavi, Machine learning for credit scoring: Improving logistic regression with non-linear decision-tree effects, European Journal of Operational Research 297 (3) (2022) 1178–1192. https://doi.org/10.1016/j.ejor.2021.06.053

[14] M. Ala'raj, M.F. Abbod, Classifiers consensus system approach for credit scoring, Knowledge-Based Systems 104 (2016) 89–105. https://doi.org/10.1016/j.knosys.2016.04.013

[15] J. Xiao, X. Zhou, Y. Zhong, L. Xie, X. Gu, D. Liu, Cost-sensitive semi-supervised selective ensemble model for customer credit scoring, Knowledge-Based Systems 189 (2020) 105118. https://doi.org/10.1016/j.knosys.2019.105118

[16] H. Qian, P. Ma, S. Gao, Y. Song, Soft reordering one-dimensional convolutional neural network for credit scoring, Knowledge-Based Systems 266 (2023) 110414. https://doi.org/10.1016/j.knosys.2023.110414

[17] M.K. Nallakaruppan, B. Balusamy, M.L. Shri, V. Malathi, S. Bhattacharyya, An Explainable AI framework for credit evaluation and analysis, Applied Soft Computing 153 (2024) 111307. https://doi.org/10.1016/j.asoc.2024.111307

[18] C.K. Chow, On optimum recognition error and reject tradeoff, IEEE Transactions on Information Theory 16 (1) (1970) 41–46. https://doi.org/10.1109/TIT.1970.1054406

[19] X.-Y. Zhang, G.-S. Xie, X. Li, T. Mei, C.-L. Liu, A survey on learning to reject, Proceedings of the IEEE 111 (2) (2023) 185–215. https://doi.org/10.1109/JPROC.2023.3238024

[20] B. Hanczar, Performance visualization spaces for classification with rejection option, Pattern Recognition 96 (2019) 106984. https://doi.org/10.1016/j.patcog.2019.106984

[21] Y. Xu, G. Kou, D. Ergu, Profit-based uncertainty estimation with application to credit scoring, European Journal of Operational Research 325 (2) (2025) 303–316. https://doi.org/10.1016/j.ejor.2025.03.007

[22] N. Khalili, A. Saleh Sedghpour, M. Jahanbakht, Z. Hosseini, From credit scoring to governed decision systems: Integrating predictive performance, uncertainty, and governance, Machine Learning and Knowledge Extraction 8 (10) (2026) 308. https://doi.org/10.3390/make8100308

[23] D. Šabanović, T. Krčmar, Z. Krpić, I. Lukić, Calibration, architecture, and distribution shift in predictive uncertainty estimation, Machine Learning and Knowledge Extraction 8 (7) (2026) 179. https://doi.org/10.3390/make8070179

[24] I.-C. Yeh, C.-H. Lien, The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients, Expert Systems with Applications 36 (2) (2009) 2473–2480. https://doi.org/10.1016/j.eswa.2007.12.020

[25] I. Brown, C. Mues, An experimental comparison of classification algorithms for imbalanced credit scoring data sets, Expert Systems with Applications 39 (3) (2012) 3446–3453. https://doi.org/10.1016/j.eswa.2011.09.033